



Week 6 Normalization

ISB02503202 - Database

Even Semester 2025/2026
3 SKS

ELEARN.UC.AC.ID



Review

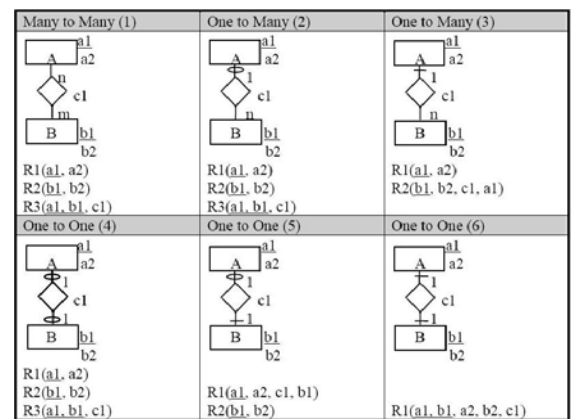
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What we already know about...

- ER Modeling
- Relational Model
- How to Convert ERD to Relational Model
-



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Terms in Relational Model

- Table (Relation)
- Functional Dependency
- Determinant
- Candidate Key
- Composite Key
- Primary Key
- Surrogate Key
- Foreign Key
- Referential Integrity Constraint
- Normal Form
- Multivalued Dependency



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5 different types of Key

- Candidate Key □ Key can determine another column in a relation
- Composite Key □ Key consisting of 2 or more column
- Primary Key □ Candidate Key selected as the primary key to identify row
- Surrogate Key □ the additional Column on the relation acts as primary key
- Foreign Key □ Key is connecting between relation

Key is combination of one or more column used to **identify** a row in a relation

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Relation (= Table*)

- Is set of some **tuples** (data)
- A model has **column** (kolom) and **row** (baris)

EmployeeNumber	FirstName	LastName	Department	Email	Phone
100	Jerry	Johnson	Accounting	JJ@somewhere.com	834-1101
200	Mary	Abemathy	Finance	MA@somewhere.com	834-2101
300	Liz	Smathers	Finance	LS@somewhere.com	834-2102
400	Tom	Caruthers	Accounting	TC@somewhere.com	834-1102
500	Tom	Jackson	Production	TJ@somewhere.com	834-4101
600	Eleanore	Caldera	Legal	EC@somewhere.com	834-3101
700	Richard	Bandalone	Legal	RB@somewhere.com	834-3102

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Characteristics Relation (Table)

- A row contained 1 record/data stored in a entity (*entity instances*)
- A column contained **attribute** of entity
- All data on the same column **has same data type**
- Each column has **unique name** (can't be the same)
- 1 cell on the table contained 1 **single/atomic value** (can't be multivalue)
- The order in the column and row doesn't matters
- No 2 rows can be the same

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Functional Dependency

- *Functional dependency* appears when the value of one attribute determines value of another attribute.
- E.g.:
 - ObjectColor $\square$ Weight
 - ObjectColor $\square$ Shape
- Read: ObjectColor determine Weight and Shape

Determinant / Dependent

- Attribute the left side on the functional dependency called **determinant**
- Attribute the right side on the functional dependency called **dependent**

ObjectColor $\square$ (Weight, Shape)
(determinant) (dependent)

Normal Form

- Relations are not the same as each other.
- Relations are categorized into normal forms based on the types of problems they address.
- Categories of normal forms:
 - 1NF
 - 2NF
 - 3NF
 - BCNF
 - 5NF
 - DK/NF

Further discussion in
Week 06.

Summary of Normalization Theory

Source of Anomaly	Normal Forms	Design Principles
Functional dependencies	1NF, 2NF, 3NF, BCNF	BCNF: Design tables so that every determinant is a candidate key.
Multivalued dependencies	4NF	4NF: Move each multivalued dependency to a table of its own.
Data constraints and oddities	5NF, DK/NF	DK/NF: Make every constraint a logical consequence of candidate keys and domains.

Normalization

Introduction

- How many tables/relations should be used?
- What fields should be present in these tables?
- What relationships are required between these tables?

The process of answering all these questions is called **database normalization**.

Common Terms in Normalization

- Relation (Table)
- Functional Dependency (FD)
 - Fully FD
 - Partly FD
- Determinant
- Dependent
- Relational Key (Key)
- Non-Key Attribute (NKA)
- Atomic Value
- Multivalued Attribute
- Composite Attribute

Functional Dependency

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Determinant / Dependent

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ObjectColor □ **(Weight, Shape)**
(determinant) (dependent)

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How to Determine Functional Dependency (FD)

	SKU	SKU_Description	Department	Buyer
1	100100	Std. Scuba Tank, Yellow	Water Sports	Pete Hansen
2	100200	Std. Scuba Tank, Magenta	Water Sports	Pete Hansen
3	100300	Std. Scuba Tank, Light Blue	Water Sports	Pete Hansen
4	100400	Std. Scuba Tank, Dark Blue	Water Sports	Pete Hansen
5	100500	Std. Scuba Tank, Light Green	Water Sports	Pete Hansen
6	100600	Std. Scuba Tank, Dark Green	Water Sports	Pete Hansen
7	101100	Dive Mask, Small Clear	Water Sports	Nancy Meyers
8	101200	Dive Mask, Med Clear	Water Sports	Nancy Meyers
9	201000	Half-dome Tent	Camping	Cindy Lo
10	202000	Half-dome Tent Vestibule	Camping	Cindy Lo
11	203000	Half-dome Tent Vestibule - Wide	Camping	Cindy Lo
12	301000	Light Fly Climbing Harness	Climbing	Jerry Martin
13	302000	Locking Carabiner, Oval	Climbing	Jerry Martin

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Answer FD (1)

- SKU □ (SKU_Description, Department, Buyer)
- SKU_Description □ (SKU, Department, Buyer)
- Buyer □ Department

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Another Example

	OrderNumber	SKU	Quantity	Price	ExtendedPrice
1	1000	201000	1	300.00	300.00
2	1000	202000	1	130.00	130.00
3	2000	101100	4	50.00	200.00
4	2000	101200	2	50.00	100.00
5	3000	100200	1	300.00	300.00
6	3000	101100	2	50.00	100.00
7	3000	101200	1	50.00	50.00

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Answer FD (2)

- (OrderNumber, SKU) □ (Quantity, Price, ExtendedPrice)
- (Quantity, Price) □ ExtendedPrice

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2 types of Functional Dependency

- **Fully** FD occurs when a **non-key attribute** is **dependent on the entire** composite candidate key, rather than just a part of it.
 - E.g.: (EmpID, ProjectID) → HoursOfWork
- **Partly** FD occurs when a **non-key attribute** is **dependent on only a proper subset** of a composite candidate key, rather than the entire key.
 - E.g.: (EmpID, ProjectID) → ProjectName

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Types of Normal Form using FD

- 1st Normal Form (1NF)
- 2nd Normal Form (2NF)
- 3rd Normal Form (3NF)
- Boyce-Codd Normal Form (BCNF)

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1st Normal Form

A relation R is said to be in 1NF if and only if all its attribute values are **atomic value**, and its **non-key attributes (if any) are not FFD** (or partly FD) on the relational key(s).

For example, a field like Address that contains a street name, a city name, and a zip code in one field cannot be said to be in 1NF because it contains three distinct values: street, city, and zip code, which should be separated into three distinct fields.

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2nd Normal Form

A relation R is called 2NF if and only if its non-key attributes (if any) are **FFD** on the relational key(s), and there are **dependencies between these non-key attributes**.

3rd Normal Form

A relation R is said to be in 3NF if and only if its non-key attributes:

- There is **no mutual dependence** (Mutually Independent)
- **FFD on relational key(s)**
- There is a **determinant that is not a relational key(s)**

Boyce-Codd Normal Form (BCNF)

A relation R is called BCNF if and only if its non-key attributes are:

- There is **no mutual dependence** (Mutually Independent)
- **FFD on relational key(s)**
- **All determinants are relational key(s)**

Steps to Determine NF

1. Determine all functional dependencies between attributes in a relation.
2. Determine all relation keys of the relation.
3. Determine key and non-key attributes.
4. Are **all non-key attributes FFD for each relation key**?
 - If NO, then the relation is in 1NF. STOP.
 - If YES, then the relation is in at least 2NF. Proceed to Step 5.

Steps to Determine NF

5. Are there **functional dependencies between non-key attributes**?
 - If YES, then the relation is in 2NF. STOP.
 - If NO, then the relation is in at least 3NF. Proceed to Step 6.
6. If the **relation has only ONE relation key**, then the relation is in BCNF. If the relation has more than one relation key, proceed to Step 7.
7. Are **all determinants determined in Step 1 relation keys**?
 - If YES, then the relation is in BCNF.
 - If NO, then the relation is only in 3NF.

Keyword

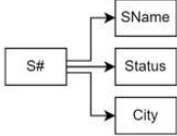
1NF = ada NKA yg Partly FD (PFD)

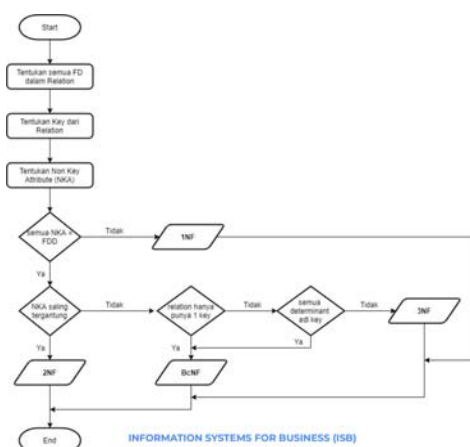
2NF = ada NKA yang saling bergantung

3NF = ada determinant yang bukan key

BCNF = semua determinant adalah key

FD Diagram

Inline	FD Diagram
STUDENT (S#, SName, Status, City) S# → SName S# → Status S# → City	



Conclusion

- The normalization process of a database is essentially the process of finding and eliminating all redundant (excessive/unnecessary) data.
- During this process, the number of fields in a table typically decreases, while the number of tables in the application increases.
- A good relational database design typically has many simple tables.

Exercise

Determine the **relational key(s)**, **normal form**, and the reason for the following relations. **Draw a FD diagram** first for each of the following relations.

#1

BRANCH (Brand_Id, City_In, Company_Id, Main_Product)

Brand_Id → City_In

Brand_Id → Company_Id

Company_Id → Main_Product

Exercise

#2

TUTORIAL (Unit_Id, Tut_No, Time, Place, Duration, Level)

Unit_Id, Tut_No → Time, Place

Time, Place → Unit_Id, Tut_No

Unit_Id, Tut_No → Duration

Unit_Id → Level

Exercise

#3

PESTY_FRUIT (Pest_Id, Season, Degree, Fruit_Id, Acidity, Colour)

Pest_Id → Season

Fruit_Id → Acidity, Colour

Pest_Id, Fruit_Id → Degree

Exercise

#4

TAXI_DRIVER (Driver_No, Age, Reg_No, Make, Model)

Driver_No → Age, Reg_No

Reg_No → Driver_No, Make, Model

Exercise

#5

ALL (U_Id, U_Name, Stream, Rating, Sem_No, Budget, Stud_Id, Status, Grade)

U_Id → U_Name, Stream

U_Name → U_Id

Sem_No → Budget

Stud_Id → Status

U_Id, Sem_No → Rating

U_Id, Sem_No, Stud_Id → Grade

THANK YOU

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